



Homework 3

OOP Fundamentals

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General

- Due to our recent success in developing simulations for Imperial Training Command, we have been tasked to create a large-scale simulation that models Stormtrooper combat. **We will begin this task by developing a simple model of Stormtrooper objects.**
- These objects are subtypes of the general Stormtrooper.
- Here is what we know about **AssaultTroopers** and **ScoutTroopers**.

Characteristic	AssaultTrooper	ScoutTrooper
Trooper ID Number	formatted as ASLT -[RN (0-9)]*[RN (10-109)]	formatted as SCT -[RN (0-9)] #[RN (10-109)]
Trooper Rank	Possible choices are Private , Lance Corporal , Corporal , or Sergeant	Possible choices are Private , Lance Corporal , Corporal , Sergeant , or Staff Sergeant
Trooper Height	Between 72 and 75 inches inclusive	Between 70 and 73 inches inclusive
Trooper Weight	3.1 lbs per inch of height	2.8 lbs per inch of height
Trooper Age	Privates are between 18 and 19 years old Lance Corporals are between 19 and 21 years old Corporals are between 20 and 25 years old Sergeants are between 26 and 32 years old Staff Sergeants are between 30 and 39 years old	

RN = randomly generated integer

Program Structure

```
▼ HW3_ProfessorGene
  > JRE System Library [JavaSE-23]
  ▼ src
    ▼ locklear.MAIN
      > Gene.java
    ▼ locklear.TRPRS
      > AssaultTrooper.java
      > ScoutTrooper.java
      > Stormtrooper.java
    ▼ locklear.TRPRS_FACTORY
      > TrooperBuilder.java
```

Create your Program Structure **EXACTLY** as shown replace my name with yours where appropriate.

Create your main method **EXACTLY** as shown replace my name with yours where appropriate.

```
public class Gene {

    public static void main(String[] args) {
        System.out.println("___Manual Building Process___\n");
        //Assault Trooper
        AssaultTrooper A1 = new AssaultTrooper("ASLT-0*0", "Private", 72, 223.2, 18);
        A1.displayAssaultTrooper();
        System.out.println();
        //Scout Trooper
        ScoutTrooper S1 = new ScoutTrooper("SCT-#0#0", "Private", 70, 196, 18);
        S1.displayScoutTrooper();
        System.out.println("\n___Automatic Building Process___\n");
        //Build and Display Random Trooper Array
        TrooperBuilder.displayAllTroopers(TrooperBuilder.buildTroopers(2, 2));
    }
}
```

Stormtrooper Class

Stormtrooper

-**String** trprID
-**String** trprType
-**String** trprRank
-**int** trprHeight
-**double** trprWeight
-**int** trprAge
+**static String** nationality = "Galactic Empire"
+**static int** stormtrooperCount = 0;

Stormtrooper()

Stormtrooper(**String** trprID)

Stormtrooper(**String** trprID, **String** trprType, **String** trprRank, **int** trprHeight, **double** trprWeight, **int** trprAge)

+toString(): **String**

+displayTrooper(): **void**

All constructors increment the **stormtrooperCount** value by 1 at instantiation.

Stormtrooper Subclasses

Stormtrooper

See Slide 4

Constructor Initialization Details

- No Argument constructors **do not** initialize any state
- All other constructors **must** initialize **all** state

AssaultTrooper

-**String** assaultCode
+**static int** assaultTroopers = 121

AssaultTrooper()
AssaultTrooper(**String** trprID, **String** trprRank,
int trprHeight, **double** trprWeight, **int** trprAge)
AssaultTrooper(**String** trprID)

+displayAssaultTrooper(): **void**

ScoutTrooper

-**String** scoutCode
+**static int** scoutTroopers = 56

ScoutTrooper()
ScoutTrooper(**String** trprID, **String** trprRank,
int trprHeight, **double** trprWeight, **int** trprAge)
ScoutTrooper(**String** trprID)

+displayScoutTrooper(): **void**

All constructors for each subclass **must utilize** their superclass constructor.

All constructors for each subclass increment their respective **assaultTrooper/scoutTroopers** value by 1 at instantiation.

Method Specification Charts

Stormtrooper Method Specification Chart

Method	Input	Processing
toString	None	Returns a String representation of the Stormtrooper object in any format.
displayTrooper	None	Prints to the console a String representation of the Stormtrooper object by utilizing the toString method in any format.

Stormtrooper Subclass Method Specification Chart

Method	Input	Processing
displayAssaultTrooper	None	Prints to the console a String representation of the AssaultTrooper object by utilizing the toString and/or displayTrooper methods of the superclass in the format shown below.
displayScoutTrooper	None	Prints to the console a String representation of the ScoutTrooper object by utilizing the toString and/or displayTrooper methods of the superclass in the format shown below.

ASSAULT TROOPER

Trpr ID: ASLT-0*0
Trpr Type: Assault
Trpr Rank: Private
Trpr Height: 72
Trpr Weight: 223.20
Trpr Age: 18
Assault Code: ASLT-0*0_121

###SCOUT TROOPER###

Trpr ID: SCT-#0#0
Trpr Type: Scout
Trpr Rank: Private
Trpr Height: 70
Trpr Weight: 196.00
Trpr Age: 18
Scout Code: SCT-#0#0_56

TrooperBuilder Class

TrooperBuilder

+static generateID(String trprType): **String**
+static generateRank(String trprType): **String**
+static generateHeightWeight(String trprType): **double[]**
+static generateAge(String trprRank): **int**
+static buildTroopers(int scout, int assault): **Stormtrooper[]**
+static displayAllTroopers(Stormtrooper [] trprs): **void**

TrooperBuilder Method Specification Chart

Method	Input	Processing
generateID	trprType	Returns a randomly generated id for the specified trprType based on the information on Slide 2
generateRank	trprType	Returns a randomly generated rank for the specified trprType based on the information on Slide 2
generateHeightWeight	trprType	Returns a randomly generated double array containing the height and weight for the specified trprType based on the information on Slide 2
generateAge	trprRank	Returns a randomly generated age based on the information on Slide 2
buildTroopers	scout assault	Returns a 1D Stormtrooper array containing the specified number of randomly generated ScoutTrooper (scout) and AssaultTrooper (assault) objects
displayAllTroopers	trprs	Calls the appropriate displayAssaultTrooper or displayScoutTrooper method for each object in the trprs array.

Expected Output

Problems @ Javadoc Declaration Console X
<terminated> Gene (19) [Java Application] C:\Users\msgth\p2\pool

___Manual Building Process___

ASSAULT TROOPER

Trpr ID: ASLT-0*0
Trpr Type: Assault
Trpr Rank: Private
Trpr Height: 72
Trpr Weight: 223.20
Trpr Age: 18
Assault Code: ASLT-0*0_121

###SCOUT TROOPER###

Trpr ID: SCT-#0#0
Trpr Type: Scout
Trpr Rank: Private
Trpr Height: 70
Trpr Weight: 196.00
Trpr Age: 18
Scout Code: SCT-#0#0_56

___Automatic Building Process___

###SCOUT TROOPER###

Trpr ID: SCT-4#4010
Trpr Type: Scout
Trpr Rank: Lance Corporal
Trpr Height: 71
Trpr Weight: 198.80
Trpr Age: 19
Scout Code: SCT-4#4010_57

###SCOUT TROOPER###

Trpr ID: SCT-0#3710
Trpr Type: Scout
Trpr Rank: Corporal
Trpr Height: 71
Trpr Weight: 198.80
Trpr Age: 20
Scout Code: SCT-0#3710_58

ASSAULT TROOPER

Trpr ID: ASLT-6*7710
Trpr Type: Assault
Trpr Rank: Private
Trpr Height: 74
Trpr Weight: 229.40
Trpr Age: 19
Assault Code: ASLT-6*7710_122

ASSAULT TROOPER

Trpr ID: ASLT-6*9210
Trpr Type: Assault
Trpr Rank: Private
Trpr Height: 74
Trpr Weight: 229.40
Trpr Age: 19
Assault Code: ASLT-6*9210_123